**The Chinese University of Hong Kong, Shenzhen**
**School of Data Science**



**MAT3007 · Homework 1**

Due: 12:00pm, June 15

**Instructions:**

- Homework problems must be carefully and clearly answered to receive full credit. Complete sentences that establish a clear logical progression are highly recommended.
- You must submit your assignment in Blackboard. Please upload a file or a zip file. The file name should be in the format **last name-first name-hw1**.
- The homework must be written in English.
- Late submission will not be graded.
- Each student **must not copy** homework solutions from another student or from any other source.
- For those questions that ask you to write MATLAB codes to solve the problem, please attach the code to the homework. You also need to clearly state (write or type) the optimal solution and the optimal value you obtained. However, you do not need to attach the outputs in the command window of MATLAB.

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**Problem 1 (25pts).** A company produces two kinds of products. A product of the first type requires 1/8 hours of assembly labor, 1/4 hours of testing, and \$1.2 worth of raw materials. A product of the second type requires 1/2 hours of assembly, 1/6 hours of testing, and \$0.9 worth of raw materials. Given the current personnel of the company, there can be at most 80 hours of assembly labor and 60 hours of testing each day. Products of the first and second type have a market value of \$8 and \$6 respectively.

(a) Formulate a linear optimization that maximizes the daily profit of the company. (For simplicity, you do not need consider the completeness of products. That is, the variables may take float values.)

(b) Write the standard form of the LP you formulated in part (a).

(c) Consider the following modification to the original problem: Suppose that up to 30 hours of overtime assembly labor can be scheduled, at a cost of \$7 per hour. Can it be easily incorporated into the linear optimization formulation and how?

(d) Solve the LP using MATLAB (for the original problem).

*Solution.*

(a) Let $x_1$ be the number of type 1 product, and $x_2$ be the number of type 2 product.

$$\begin{array}{ll} \text{maximize} & (8-1.2)x_1 + (6-0.9)x_2 \\ \text{s.t.} & \frac{1}{8}x_1 + \frac{1}{2}x_2 \leq 80 \\ & \frac{1}{4}x_1 + \frac{1}{6}x_2 \leq 60 \\ & x_1, x_2 \geq 0 \end{array}$$

(b) The standard form is as follows.

$$\begin{array}{ll} \text{minimize} & -(8-1.2)x_1 - (6-0.9)x_2 \\ \text{s.t.} & \frac{1}{8}x_1 + \frac{1}{2}x_2 + s_1 = 80 \\ & \frac{1}{4}x_1 + \frac{1}{6}x_2 + s_2 = 60 \\ & x_1, x_2, s_1, s_2 \geq 0 \end{array}$$

(c) Let $x_3$ be the number of overtime hours. We can form the following LP.

$$\begin{array}{ll} \text{maximize} & (8-1.2)x_1 + (6-0.9)x_2 - 7x_3 \\ \text{s.t.} & \frac{1}{8}x_1 + \frac{1}{2}x_2 \leq 80 + x_3 \\ & \frac{1}{4}x_1 + \frac{1}{6}x_2 \leq 60 \\ & x_3 \leq 30 \\ & x_1, x_2, x_3 \geq 0 \end{array}$$

(d) The optimal objective value is 1700. The optimal solutions are $x_1 = 160, x_2 = 120$.

```
cvx_begin
variables x1 x2
minimize -(8-1.2)*x1 - (6-0.9)*x2
subject to
1/8*x1 + 1/2*x2 <= 80;
1/4*x1 + 1/6*x2 <= 60;
x1 >= 0;
x2 >= 0;
cvx_end
```

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**Problem 2 (25pts).** The China Railroad Ministry is in the process of planning relocations of freight cars among 5 regions of the country to get ready for the fall harvest. Table1a shows the cost of moving a car between each pair of regions. Table1b shows the current number of cars in each region and the number needed for harvest shipping.
Write down a linear optimization to compute the least costly way to move the cars such us the need is met. Then solve the problem using MATLAB.

*Solution.*

| From/To | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 | – | 20 | 13 | 11 | 28 |
| 2 | 20 | – | 18 | 8 | 46 |
| 3 | 13 | 18 | – | 9 | 27 |
| 4 | 11 | 8 | 9 | – | 20 |
| 5 | 28 | 46 | 27 | 20 | – |

(a) Cost of moving a car

| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Present | 120 | 330 | 400 | 400 | 600 |
| Need | 150 | 200 | 600 | 200 | 400 |

(b) Number of current and needed cars

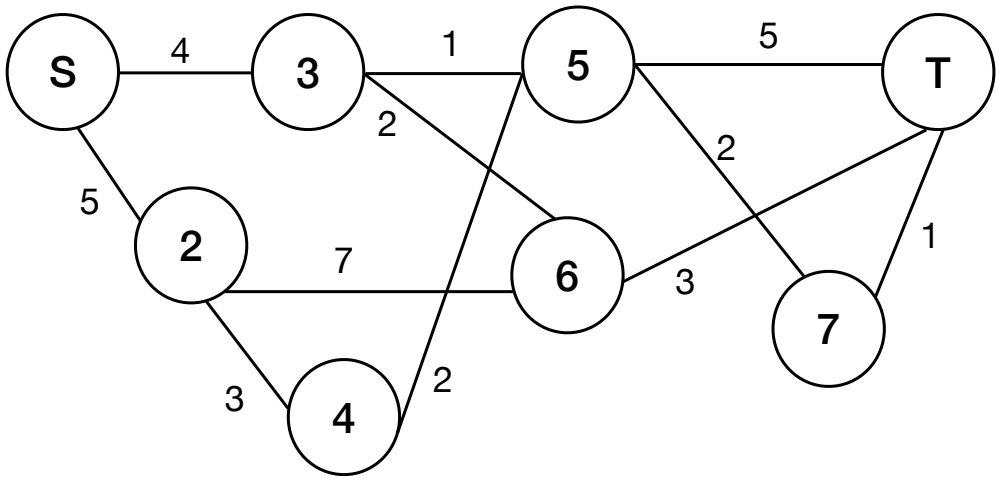



Figure 1: The graph of the shortest path problem

Let $x_{ij}$ denote the number of cars moving from place $i$ to $j$.

$$
\begin{array}{ll}
\text{minimize} & \sum_{i=1}^{5}\sum_{j=1}^{5} c_{ij}x_{ij} \\
\text{s.t.} & \sum_{i\neq 1} x_{i1} - \sum_{j\neq 1} x_{1j} \geq 150 - 120 \\
& \sum_{i\neq 2} x_{i2} - \sum_{j\neq 2} x_{2j} \geq 200 - 330 \\
& \sum_{i\neq 3} x_{i3} - \sum_{j\neq 3} x_{3j} \geq 600 - 400 \\
& \sum_{i\neq 4} x_{i4} - \sum_{j\neq 4} x_{4j} \geq 200 - 400 \\
& \sum_{i\neq 5} x_{i5} - \sum_{j\neq 5} x_{5j} \geq 400 - 600 \\
& x_{ij} \geq 0 \quad \forall i, j
\end{array}
$$

The optimal solutions are $x_{41} = 30, x_{43} = 200, x_{24} = 30,$, with the rest $x_{ij} = 0$. The optimal objective value is 2370.

```
c = [0 20 13 11 28;
20 0 18 8 46;
13 18 0 9 27;
11 8 9 0 20;
```

```
28 46 27 20 0];

cvx_begin
variables x(5,5)

minimize sum(sum(c.*x))
subject to

x(2,1)+x(3,1)+x(4,1)+x(5,1)-x(1,2)-x(1,3)-x(1,4)-x(1,5)>=150-120;
x(1,2)+x(3,2)+x(4,2)+x(5,2)-x(2,1)-x(2,3)-x(2,4)-x(2,5)>=200-330;
x(1,3)+x(2,3)+x(4,3)+x(5,3)-x(3,1)-x(3,2)-x(3,4)-x(3,5)>=600-400;
x(1,4)+x(2,4)+x(3,4)+x(5,4)-x(4,1)-x(4,2)-x(4,3)-x(4,5)>=200-400;
x(1,5)+x(2,5)+x(3,5)+x(4,5)-x(5,1)-x(5,2)-x(5,3)-x(5,4)>=400-600;
x >= zeros(5,5);
cvx_end
```

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**Problem 3 (20pts).** Write a MATLAB code to use linear optimization to solve the shortest path problem. Suppose the input of the problem will be an $n \times n$ matrix of $W$, where $w_{ij}$ is the length of the path from $i$ to $j$ (in general, $w_{ij}$ does not necessarily equal $w_{ji}$). In our implementation, we always use 1 to denote the source node (the $s$ node in the lecture slides), and $n$ to denote the terminal node (the $t$ node in the lecture slides). In addition, we assume for any $i$ and $j$, there is a path, i.e., the set of $E$ is all pairs of nodes. This is without loss of generality since one can set $w_{ij}$ to be an extremely large number if there is no edge between $i$ and $j$, effectively eliminating it from consideration.
After writing the code, you are asked to solve the concrete problem in the lecture slides, with the given labeling shown in Figure 1. Basically, you need to input the $W$ matrix for this case, then solve it, and then report your solution (the optimal path).

*Solution.*
The optimal solutions are $x(1,3) = x(3,5) = x(5,7) = x(7,8) = 1$, with the rest $x(i,j) = 0$. The optimal objective value is 8.

```
M = 100;
W = [M, 5, 4, M, M, M, M, M;
5, M, M, 3, M, 7, M, M;
4, M, M, M, 1, 2, M, M;
M, 3, M, M, 2, M, M, M;
M, M, 1, 2, M, M, 2, 5;
M, 7, 2, M, M, M, M, 3;
M, M, M, M, 2, M, M, 1;
M, M, M, M, 5, 3 ,1, M];
[n, m] = size(W);
cvx_begin
variables x(n,n)
minimize sum(sum(W .* x))
subject to
sum(x(1,:)) == 1;
sum(x(:,n)) == 1;
sum(x(2:n-1,:), 2) - sum(x(:,2:n-1))' == 0;
0 <= x <= 1;
cvx_end
```

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**Problem 4 (30pts).** Reformulate the following problems as linear programming problems

(a)

$$\begin{array}{ll} \min & 3x + 4|y - x| \\ \text{s.t.} & |x + 1| + |y| \leq 6 \end{array}$$

where $x, y \in \mathbb{R}$.

(b)

$$\begin{array}{ll} \min & c^\top x + f(d^\top x) \\ \text{s.t.} & Ax \geq b \end{array}$$

where $x, c, d \in \mathbb{R}^n, A \in \mathbb{R}^{m\times n}, b \in \mathbb{R}^m$, $f(\alpha) = \max\{\alpha, 2, 3\alpha - 5\}$ for $\alpha \in \mathbb{R}$,

(c)

$$\begin{array}{ll} \min & c^\top x \\ \text{s.t.} & \|Ax - b\|_\infty \leq \delta \\ & x \geq 0 \end{array}$$

where $x \in \mathbb{R}^n, A \in \mathbb{R}^{m\times n}, b \in \mathbb{R}^m, ||y||_\infty = \max_i |y_i|$ for $y \in \mathbb{R}^n$.

*Solution.*

(a) Introduce new variable $z \geq |x - y|$, which is equivalent to $z \geq x - y, z \geq y - x$. $|x+1|+|y| \leq 6$ is equivalent to $x + 1 + y \leq 6$, $x + 1 - y \leq 6$, $-x - 1 + y \leq 6$, $-x - 1 - y \leq 6$. So the original problem can be reformulated as

$$\begin{array}{ll} \min_{x,y,z} & 3x + 4z \\ \text{s.t.} & x + 1 + y \leq 6 \\ & x + 1 - y \leq 6 \\ & -x - 1 + y \leq 6 \\ & -x - 1 - y \leq 6 \\ & y - x \leq z \\ & x - y \leq z \end{array}$$

(b) Let $y = f(d^\top x)$, then $y \geq d^\top x, y \geq 2, y \geq 3d^\top x - 5$. So the problem can be reformualted as

$$\begin{array}{ll} \min & c^\top x + y \\ \text{s.t.} & Ax \geq b \\ & y \geq d^\top x \\ & y \geq 2 \\ & y \geq 3d^\top x - 5 \end{array}$$

(c) Let the rows of $A$ to be $a_1, a_2, \ldots, a_m$, then $\|Ax - b\|_\infty \leq \delta$ is equivalent to $|a_i x - b| \leq \delta, i = 1, 2, \ldots m$. So the problem is equivalent to

$$\begin{array}{ll} \min & c^\top x \\ \text{s.t.} & a_i x - b \leq \delta, i = 1, 2, \ldots m \\ & -a_i x + b \leq \delta, i = 1, 2, \ldots m \\ & x \geq 0 \end{array}$$

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